

L.
Continuity and Irrational Numbers

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*Dedicated to his beloved father,
the
Privy Councillor, Professor, Dr. jur. Julius Levin Ulrich Dedekind
in Braunschweig, on the occasion of the fiftieth anniversary of his service
on 26 April 1872.*

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Preface

The considerations which form the subject of this little writing date from the autumn of 1858. At that time, as professor at the Federal Polytechnic in Zurich, I found myself for the first time in the position of having to lecture on the elements of the differential calculus, and in doing so I felt, more sharply than ever before, the lack of a truly scientific foundation for arithmetic. In the concept of the approach of a variable magnitude to a fixed limiting value, and especially in the proof of the theorem that every magnitude which continually increases, but not beyond all bounds, must certainly approach a limiting value, I took refuge in geometric evidences. Even now I consider such recourse to geometric intuition, in the first instruction in the differential calculus, to be extraordinarily useful from the didactic standpoint, indeed indispensable, if one does not wish to lose too much time. But no one will deny that this manner of introducing the differential calculus can lay no claim to scientificity. For me at that time this feeling of dissatisfaction was so overwhelming that I formed the firm resolution to reflect until I had found a purely arithmetical and wholly rigorous foundation for the principles of infinitesimal analysis. It is so often said that the differential calculus is concerned with continuous magnitudes, and yet nowhere is an explanation of this continuity given; and even the strictest presentations of the differential calculus do not base their proofs on continuity, but either, with more or less awareness, appeal to geometric ideas, or to ideas occasioned by geometry, or else rely on theorems which have themselves never been proved purely arithmetically. Among these belongs, for example, the theorem mentioned above; and a more exact investigation convinced me that this theorem, or indeed any theorem equivalent to it, can in a certain sense be regarded as a sufficient foundation for infinitesimal analysis. It remained only to discover its proper origin in the elements of arithmetic and thereby at the same time to obtain a real definition of the essence of continuity. This I succeeded in doing on 24 November 1858, and a few days later I communicated the result of my reflections to my dear friend Durege, which led to a long and lively conversation. Later I did explain these thoughts on a scientific foundation of arithmetic to one or another of my students, and also here in Braunschweig gave a lecture on this subject in the scientific society of professors; but I could not quite bring myself to an actual publication, first because the presentation is not wholly easy, and also because the subject is so little fruitful. Nevertheless I had already half thought of choosing this theme as the subject of this occasional writing when, a few days ago, on 14 March, the paper "The Elements of the Theory of Functions" by E. Heine (Crelle's Journal,

vol. 74) came into my hands through the kindness of its much-honoured author and strengthened me in my decision. In substance I am indeed in complete agreement with the contents of that writing, as could not be otherwise; but I will frankly admit that my presentation seems to me simpler in form and to bring out the real central point more precisely. And while I am writing this preface (20 March 1872), I receive the interesting paper “On the Extension of a Theorem from the Theory of Trigonometric Series” by G. Cantor (*Mathematische Annalen* of Clebsch and Neumann, vol. 5), for which I offer the acute author my best thanks. As I find on rapid reading, the axiom in §2 of that paper, apart from the external form of its clothing, agrees completely with what I designate below in §3 as the essence of continuity. But what benefit may be afforded by the distinction, even if only conceptual, of real numerical magnitudes of still higher kind, I cannot yet recognize, precisely in view of my conception of the domain of real numbers as complete in itself.

§1. Properties of the rational numbers

The development of the arithmetic of the rational numbers is assumed here; nevertheless I consider it good to emphasize a few principal points without discussion, merely in order to indicate from the outset the standpoint which I adopt in what follows. I regard the whole of arithmetic as a necessary, or at least natural, consequence of the simplest arithmetical act, that of counting; and counting itself is nothing other than the successive creation of the infinite series of positive whole numbers, in which each individual is defined by the one immediately preceding it. The simplest act is the transition from an individual already created to the next one to be created. The chain of these numbers already forms, in itself, an exceedingly useful aid for the human mind, and it offers an inexhaustible wealth of remarkable laws to which one arrives by the introduction of the four fundamental arithmetical operations. Addition is the combination, into a single act, of any repetition whatever of the above simplest act; and from it multiplication arises in the same way. While these two operations are always executable, the inverse operations, subtraction and division, show only a restricted admissibility. Whatever may have been the immediate occasion, whatever comparisons or analogies with experiences and intuitions may have led to it, let that be left aside; enough, precisely this restriction in the executability of the indirect operations has each time become the real cause of a new act of creation. Thus the negative and fractional numbers have been created by the human mind, and in the system of all rational numbers an instrument of infinitely greater perfection has been obtained. This system, which I shall denote by R , possesses above all a completeness and closure which I have elsewhere¹ designated as the characteristic mark of a number field, and which consists in this: that the four fundamental operations with any two individuals in R are always executable, that is, that the result is always again a determinate individual in R , if one excepts the single case of division by the number zero.

For our present purpose, however, another property of the system R is still more important; it can be expressed by saying that the system R forms a well-ordered, one-dimensional domain extending infinitely in two opposite directions. What is meant by this is sufficiently indicated by the choice of expressions borrowed from geometric ideas; for that very reason it is all the more necessary to emphasize the corresponding purely arithmetical peculiarities, so that not even the appearance remains that arithmetic needs such ideas foreign to it.

If it is to be expressed that the signs a and b denote one and the same rational number, one writes both $a = b$ and $b = a$. The difference of two rational numbers a, b shows itself in this: that the difference $a - b$ has either a positive or a negative value. In the first case a is called greater than b , and b less than a , which is also indicated by the signs $a > b$, $b < a$.² Since in the second

¹Lectures on Number Theory by P. G. Lejeune Dirichlet. Second edition. §159.

²Thus in what follows the so-called “algebraic” greater-than and less-than relation is always meant, unless the word “absolute” is added.

case $b - a$ has a positive value, one has $b > a$, $a < b$. With respect to this double possibility in the manner of difference, the following laws hold.

- I. If $a > b$ and $b > c$, then $a > c$. Whenever a, c are two different, or unequal, numbers, and b is greater than one and less than the other, we shall express this briefly, without fear of the echo of geometric ideas, by saying that b lies between the two numbers a, c .
- II. If a, c are two different numbers, then there are always infinitely many different numbers b which lie between a, c .
- III. If a is a fixed number, then all numbers of the system R fall into two classes A_1 and A_2 , each of which contains infinitely many individuals. The first class A_1 contains all numbers a_1 which are $< a$, and the second class A_2 contains all numbers a_2 which are $> a$. The number a itself may be assigned at will to the first or to the second class, and is then correspondingly the greatest number of the first class or the least number of the second class. In every case the decomposition of the system R into the two classes A_1, A_2 is of such a kind that every number of the first class A_1 is less than every number of the second class A_2 .

§2. Comparison of the rational numbers with the points of a straight line

The properties of the rational numbers just emphasized recall the mutual relations of position among the points of a straight line L . If the two opposite directions existing in it are distinguished as “right” and “left”, and if p, q are two different points, then either p lies to the right of q , and at the same time q to the left of p , or conversely q lies to the right of p , and at the same time p to the left of q . A third case is impossible if p, q are really different points. With respect to this difference of position, the following laws hold.

- I. If p lies to the right of q , and q again to the right of r , then p also lies to the right of r ; and one says that q lies between the points p and r .
- II. If p, r are two different points, then there are always infinitely many points q which lie between p and r .
- III. If p is a fixed point in L , then all points in L fall into two classes, P_1, P_2 , each of which contains infinitely many individuals. The first class P_1 contains all the points p_1 which lie to the left of p , and the second class P_2 contains all the points p_2 which lie to the right of p . The point p itself may be assigned at will to the first or to the second class. In every case the decomposition of the straight line L into the two classes, or pieces, P_1, P_2 is of such a kind that every point of the first class P_1 lies to the left of every point of the second class P_2 .

This analogy between the rational numbers and the points of a line is known to become an actual connection when a fixed origin, or zero point, o and a fixed unit of length for the measurement of segments are chosen on the line. With the aid of the latter, a corresponding length can be constructed for every rational number a ; and if one lays this length off from the point o to the right or to the left on the line according as a is positive or negative, one obtains a determinate endpoint p , which can be designated as the point corresponding to the number a . The point o corresponds to the rational number zero. In this way to every rational number a , that is, to every individual in R , there corresponds one and only one point p , that is, one individual in L . If the two points p, q correspond respectively to the two numbers a, b , and if $a > b$, then p lies to the right of q . To the laws I, II, III of the preceding paragraph there correspond completely the laws I, II, III of the present one.

§3. Continuity of the straight line

Of the greatest importance now is the fact that in the straight line L there are infinitely many points which correspond to no rational number. For if the point p corresponds to the rational number a , then, as is known, the length op is commensurable with the unchangeable unit of length used in the construction; that is, there exists a third length, a so-called common measure, of which these two lengths are integral multiples. But already the ancient Greeks knew and proved that there are lengths which are incommensurable with a given unit of length, for example the diagonal of the square whose side is the unit of length. If one lays off such a length on the line from the point o , one obtains an endpoint which corresponds to no rational number. Since it is moreover easy to prove that there are infinitely many lengths incommensurable with the unit of length, we may assert: the straight line L is infinitely richer in point-individuals than the domain R of rational numbers is in number-individuals.

If one now wishes, as is surely desirable, to pursue all phenomena in the straight line arithmetically as well, the rational numbers do not suffice; and it therefore becomes unavoidably necessary to refine substantially the instrument R , which had been constructed through the creation of the rational numbers, by a creation of new numbers of such a kind that the domain of numbers gains the same completeness, or, as we shall immediately say, the same continuity, as the straight line.

The preceding considerations are so well known and familiar to everyone that many will regard their repetition as quite superfluous. Nevertheless I considered this recapitulation necessary in order properly to prepare the main question. For the customary introduction of irrational numbers attaches itself directly to the concept of extensive magnitudes — which itself is nowhere rigorously defined — and explains number as the result of measuring one such magnitude by a second magnitude of the same kind.³ Instead I demand that arithmetic shall develop out of itself. That such connections with non-arithmetical ideas gave the immediate occasion for extending the concept of number may in general be admitted, although this was decidedly not the case in the introduction of the complex numbers; but in this there is certainly no reason to include these heterogeneous considerations themselves in arithmetic, in the science of numbers. Just as the negative and fractional rational numbers are produced by a free creation, and just as the laws of calculation with these numbers must and can be reduced to the laws of calculation with positive whole numbers, so one must strive for the irrational numbers also to be defined completely by the rational numbers alone. Only the question “How?” remains.

The above comparison of the domain R of rational numbers with a line has led to the recognition of the gappiness, incompleteness, or discontinuity of the former, while we ascribe completeness, gaplessness, or continuity to the line. In what, then, does this continuity actually consist? Everything must be contained in the answer to this question, and only through it will one obtain a scientific basis for the investigation of all continuous domains. Nothing is gained, of course, by vague talk of uninterrupted connection in the smallest parts; what matters is to indicate a precise characteristic of continuity which can be used as a basis for actual deductions. For a long time I reflected on this in vain, but at last I found what I was seeking. Different people will perhaps judge this discovery differently; nevertheless I believe that most will find its content very trivial. It consists in the following. In the preceding paragraph attention was drawn to the fact that every point p of the line brings about a decomposition of it into two pieces of such a kind that every point of one piece lies to the left of every point of the other. I now find the essence of continuity in the converse, hence in the following principle:

“If all points of the straight line fall into two classes of such a kind that every point of the first class lies to the left of every point of the second class, then there exists

³The apparent advantage of generality in this definition of number vanishes at once when one thinks of complex numbers. Conversely, in my view, the concept of the ratio between two magnitudes of the same kind can be developed clearly only after the irrational numbers have already been introduced.

one and only one point which brings about this division of all points into two classes, this cutting of the line into two pieces.”

As already said, I believe I am not mistaken in assuming that everyone will immediately concede the truth of this assertion; most of my readers will be much disappointed to hear that the secret of continuity is supposed to be revealed by this triviality. To this I add the following. It is very welcome to me if everyone finds the above principle so evident and so in agreement with his ideas of a line; for I am unable to produce any proof of its correctness, and no one is able to do so. The assumption of this property of the line is nothing but an axiom, by which we first recognize continuity in the line, by which we think continuity into the line. If space has any real existence at all, it need not necessarily be continuous; countless of its properties would remain the same even if it were discontinuous. And if we knew for certain that space was discontinuous, still nothing could prevent us, if we pleased, from making it continuous in thought by filling its gaps. This filling, however, would consist in a creation of new point-individuals and would have to be carried out according to the above principle.

§4. Creation of the irrational numbers

The last words already indicate sufficiently the manner in which the discontinuous domain R of rational numbers must be completed to a continuous one. In §1 it was emphasized (III) that every rational number a brings about a decomposition of the system R into two classes A_1, A_2 of such a kind that every number a_1 of the first class A_1 is less than every number a_2 of the second class A_2 . The number a is either the greatest number of the class A_1 or the least number of the class A_2 . Now if any division whatever of the system R into two classes A_1, A_2 is given which possesses only the characteristic property that every number a_1 in A_1 is less than every number a_2 in A_2 , then for brevity we shall call such a division a cut and denote it by (A_1, A_2) . We can then say that every rational number a brings about a cut, or rather two cuts, which however we do not wish to regard as essentially different. This cut has in addition the property that either among the numbers of the first class there exists a greatest, or among the numbers of the second class there exists a least. Conversely, if a cut also possesses this property, then it is brought about by this greatest or least rational number.

But one easily convinces oneself that there also exist infinitely many cuts which are not brought about by rational numbers. The nearest example is the following.

Let D be a positive whole number, but not the square of a whole number; then there exists a positive whole number λ such that

$$\lambda^2 < D < (\lambda + 1)^2.$$

If one takes into the second class A_2 every positive rational number a_2 whose square is $> D$, but into the first class A_1 all other rational numbers a_1 , then this division forms a cut (A_1, A_2) , that is, every number a_1 is less than every number a_2 . For if $a_1 = 0$ or is negative, then for that reason alone a_1 is less than every number a_2 , since by definition the latter is positive. If however a_1 is positive, then its square is $< D$, and consequently a_1 is less than every positive number a_2 whose square is $> D$.

This cut, however, is brought about by no rational number. To prove this it must first of all be shown that there is no rational number whose square is equal to D . Although this is known from the first elements of number theory, the following indirect proof may nevertheless find a place here. If there is a rational number whose square is equal to D , then there also exist two positive whole numbers t, u satisfying the equation

$$t^2 - Du^2 = 0,$$

and one may assume that u is the least positive whole number with the property that its square is transformed, by multiplication by D , into the square of a whole number t . Since now evidently

$$\lambda^2 u^2 < t^2 < (\lambda + 1)^2 u^2,$$

the number

$$u' = t - \lambda u$$

is a positive whole number and indeed less than u . If one further sets

$$t' = Du - \lambda t,$$

then t' is likewise a positive whole number, and one obtains

$$t'^2 - Du'^2 = (\lambda^2 - D)(t^2 - Du^2) = 0,$$

which contradicts the assumption about u .

Accordingly, the square of every rational number x is either $< D$ or $> D$. From this it follows easily that there is neither a greatest number in the class A_1 nor a least number in the class A_2 . For if one sets

$$y = x \cdot \frac{x^2 + 3D}{3x^2 + D},$$

then

$$y - x = \frac{2x(D - x^2)}{3x^2 + D}$$

and

$$y^2 - D = \frac{(x^2 - D)^3}{(3x^2 + D)^2}.$$

If here for x one takes a positive number from the class A_1 , then $x^2 < D$, and consequently $y > x$ and $y^2 < D$; hence y likewise belongs to the class A_1 . But if for x one takes a number from the class A_2 , then $x^2 > D$, and consequently $y < x$, y is positive, and $y^2 > D$; hence y likewise belongs to the class A_2 . Therefore this cut is brought about by no rational number.

In this property, that not all cuts are brought about by rational numbers, consists the incompleteness or discontinuity of the domain R of all rational numbers.

Now whenever a cut (A_1, A_2) is present which is brought about by no rational number, we create a new, an irrational number α , which we regard as completely defined by this cut (A_1, A_2) . We shall say that the number α corresponds to this cut, or that it brings about this cut. Thus from now on there corresponds to every determinate cut one and only one determinate rational or irrational number; and we regard two numbers always, and only then, as different or unequal when they correspond to essentially different cuts.

In order now to obtain a basis for the ordering of all real numbers, that is, of all rational and irrational numbers, we must first investigate the relations between any two cuts (A_1, A_2) and (B_1, B_2) which are brought about by any two numbers α and β . Evidently a cut (A_1, A_2) is already completely given when one of the two classes, for example the first A_1 , is known, since the second A_2 consists of all rational numbers not contained in A_1 ; and the characteristic property of such a first class A_1 lies in this: that if the number a_1 is contained in it, then all numbers smaller than a_1 are also contained in it. Comparing now two such first classes A_1, B_1 with one another, it can happen, first, that they are completely identical, that is, that every number a_1 contained in A_1 is also contained in B_1 , and every number b_1 contained in B_1 is also contained in A_1 . In this case A_2 is then necessarily identical with B_2 , and the two cuts are completely identical, which we indicate symbolically by $\alpha = \beta$ or $\beta = \alpha$.

But if the two classes A_1, B_1 are not identical, then in one of them, for example in A_1 , there is a number $a'_1 = b'_2$ which is not contained in the other, B_1 , and which consequently is found in

B_2 . Therefore all numbers b_1 contained in B_1 are certainly smaller than this number $a'_1 = b'_2$, and consequently all numbers b_1 are also contained in A_1 .

Now, second, if this number a'_1 is the only one in A_1 which is not contained in B_1 , then every other number a_1 contained in A_1 is contained in B_1 and consequently is smaller than a'_1 ; that is, a'_1 is the greatest among all numbers a_1 . Thus the cut (A_1, A_2) is brought about by the rational number $\alpha = a'_1 = b'_2$. Of the other cut (B_1, B_2) we already know that all numbers b_1 in B_1 are also contained in A_1 and are smaller than the number $a'_1 = b'_2$, which is contained in B_2 . Every other number b_2 contained in B_2 must however be greater than $a'_1 = b'_2$, because otherwise it would also be smaller than a'_1 , and hence would be contained in A_1 and consequently also in B_1 . Thus b'_2 is the least among all numbers contained in B_2 , and consequently the cut (B_1, B_2) also is brought about by the same rational number $\beta = b'_2 = a'_1 = \alpha$. The two cuts are therefore only inessentially different.

But, third, if in A_1 there are at least two different numbers $a'_1 = b'_2$ and $a''_1 = b''_2$ which are not contained in B_1 , then there are also infinitely many such numbers, because all the infinitely many numbers lying between a'_1 and a''_1 (§1. II) are evidently contained in A_1 but not in B_1 . In this case we also call the numbers α and β corresponding to these two essentially different cuts (A_1, A_2) and (B_1, B_2) different from one another; and specifically we say that α is greater than β , and that β is less than α , which we express symbolically both by $\alpha > \beta$ and by $\beta < \alpha$. It should be emphasized that this definition coincides completely with the earlier one when both numbers α, β are rational.

The remaining possible cases are these. Fourth, if in B_1 there is one and only one number $b'_1 = a'_2$ which is not contained in A_1 , then the two cuts (A_1, A_2) and (B_1, B_2) are only inessentially different and are brought about by one and the same rational number $\alpha = a'_2 = b'_1 = \beta$. But, fifth, if in B_1 there are at least two different numbers which are not contained in A_1 , then $\beta > \alpha$, $\alpha < \beta$.

Since this exhausts all cases, it follows that of two different numbers necessarily one must be the greater and the other the smaller, which contains two possibilities. A third case is impossible. This did indeed already lie in the choice of the comparative (greater, smaller) to designate the relation between α, β ; but that choice is only now justified after the fact. Precisely in such investigations one must guard oneself most carefully, even with the best will to be honest, against being led by a premature choice of expressions borrowed from other already developed ideas into making illicit transfers from one domain to another.

If one now considers once more exactly the case $\alpha > \beta$, it follows that the smaller number β , if it is rational, certainly belongs to the class A_1 . For since in A_1 there is a number $a'_1 = b'_2$ which belongs to the class B_2 , the number β , whether it is the greatest number in B_1 or the least number in B_2 , is certainly $\leq a'_1$ and consequently contained in A_1 . Similarly it follows from $\alpha > \beta$ that the greater number α , if it is rational, certainly belongs to the class B_2 , since $\alpha \geq a'_1$. Combining the two considerations, one obtains the following result: if a cut (A_1, A_2) is brought about by the number α , then any rational number belongs to the class A_1 or to the class A_2 according as it is smaller or greater than α ; if the number α itself is rational, then it may belong to either class.

Finally the following also results from this. If $\alpha > \beta$, so that there are infinitely many numbers in A_1 which are not contained in B_1 , then there are also infinitely many such numbers which at the same time are different from α and from β . Every such rational number c is $< \alpha$, because it is contained in A_1 , and at the same time it is $> \beta$, because it is contained in B_2 .